

BRIGHTCON 2026 · AALBORG · 24 SEPTEMBER 2026

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Eaternity · Zürich



LCA Interface



PART 1 · FIREFLY · 1 / 6

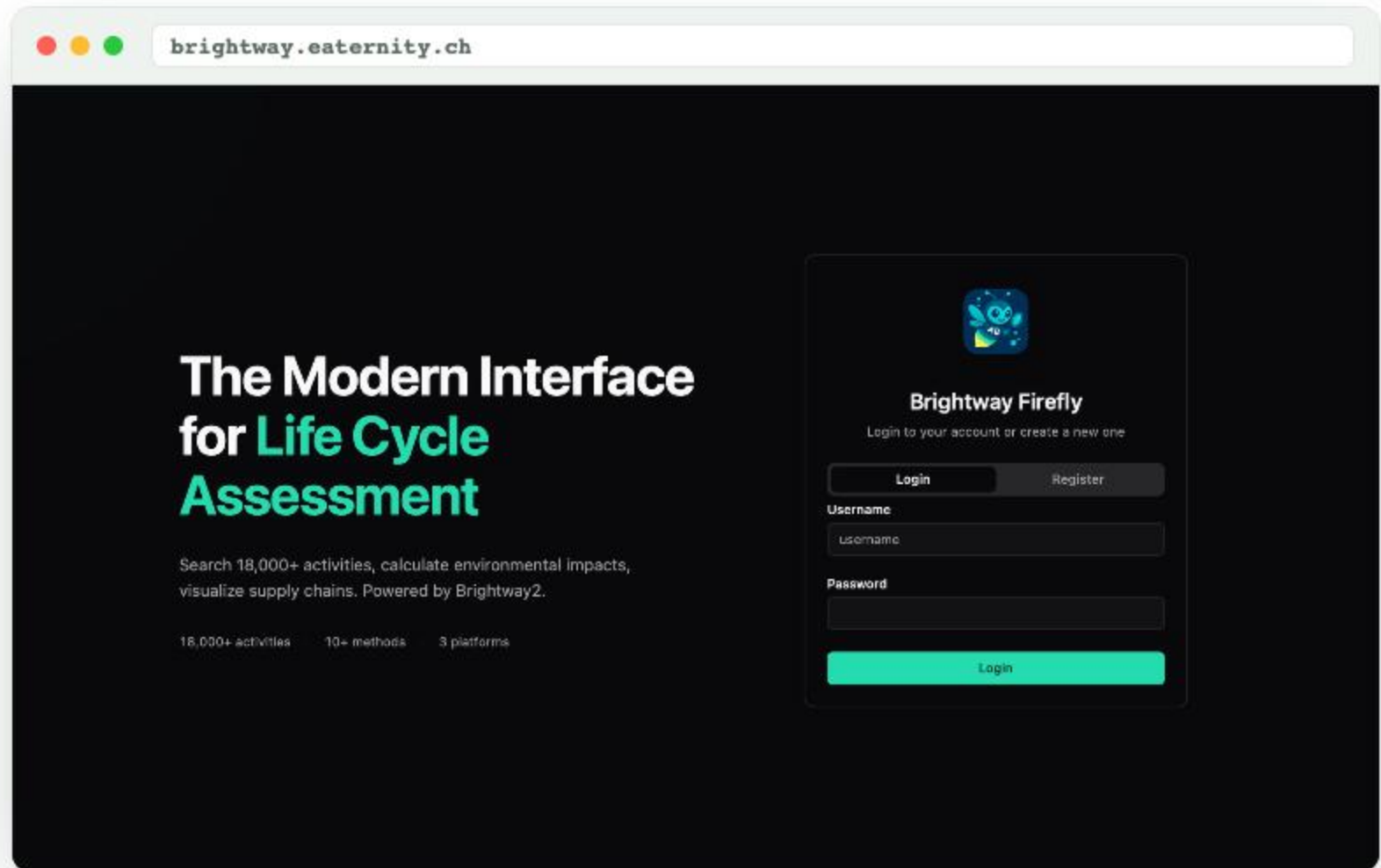
Firefly

Web-based

log in at brightway.eaternity.ch — or download it for macOS, Windows, Linux

Glossary first

say what you mean; the inventory follows



Works with your database

Bring what you have

a Brightway package or an ILCD zip from openLCA or SimaPro — from a file or straight from a URL

No lock-in

take a project or a single database back out: Brightway, Excel, openLCA, ILCD, SimaPro — with dependencies and impact methods

The databases you already use

ecoinvent, Agribalyse, USLCI, ELCD — import the package you are licensed for; bafu-2026 is shared as a Context Database to clone

Create New Project

Start a new project or restore one from a backup.

Start a new project



Empty



Remote



Import

Restore from a backup



Restore

Create a project from a Brightway data package (.bw2package) or an ILCD .zip package

Source

Upload a file

From a URL

Package File

.bw2package · ILCD .zip — the format is detected from the file extension

Export / Backup

The export runs in the background and you download the finished file.

Scope

Whole project

Format

Backup (.tar.gz)



BW2Package (Brightway)

Excel (bw2io layout)

openLCA (JSON-LD zip)

ILCD (zip)

SimaPro (CSV)

Exact restore of everything — databases, methods, parameters, setups.

Start export

Search — glossary first

Your words become glossary terms first

concept · geography · exclusion · boundary

Inventories ranked by match — and by quality

one mark per term: carried, left out, declared as left out; the quality score breaks ties

Embedding-based, so it works across languages and vocabularies

meaning, not keywords — 14 international vocabularies behind it

brightway.eaternity.ch/browser

market for wheat grain, x, CH, dried without transport

Glossary terms

wheat grain
 CONCEPT from "wheat grain dried"
 closest glossary term, not an exact match
 GLOSSARY ecoinvent:0201ee86-f71b-4457-8fe8-79f3ed720b07 wheat grain

dried
 CONCEPT
 from 'dried', not named by wheat grain
 GLOSSARY foodon:FOODON_00004723 dried

Switzerland
 GEOGRAPHY from "CH"
 GLOSSARY agrovoc:c_7558 Switzerland

transport
 EXCLUSION
 GLOSSARY agrovoc:c_7874 transport

market
 BOUNDARY

Inventories · 20 of 176

dried roughage store, non ventilated
 ch bafu-2026/585ea5...e9d6a3 QUALITY
 wheat grain dried Switzerland transport market

dried roughage store, cold-air dried, conventional
 ch bafu-2026/370544...c9edc6 QUALITY
 wheat grain dried Switzerland transport market

dried roughage store, non ventilated, operation
 ch bafu-2026/254143...83b53e QUALITY
 wheat grain dried Switzerland transport market

dried roughage store, air dried, solar
 ch bafu-2026/39397a...0b6d71 QUALITY
 wheat grain dried Switzerland transport market

xx dried roughage store, air dried, solar, operation
 ch bafu-2026/fd0ed3...57ab75 QUALITY
 wheat grain dried Switzerland transport market

dried roughage store, cold-air dried, conventional, operation
 ch bafu-2026/ffcb31...cc45a4 QUALITY
 wheat grain dried Switzerland transport market

sawnwood, board, hardwood, dried (u=20%), planed, at sawmill
 ch bafu-2026/0f2fde...7be172 QUALITY
 wheat grain dried Switzerland transport market

sawnwood, hardwood, raw, dried (u=10%), at regional storage

WHY GLOSSARY FIRST

Your intent outlives any database

Say what you mean

stored as glossary terms — never as a key into one database version

Matched across every database

the best inventory you have, found for you

Upgrades itself

a better inventory arrives → the match is re-scored

brightway.eaternity.ch/browser · glossary

Browser / V2

Search... **apple** 96863 / 101136 activities

All terms Only terms in this project

766 terms match (name, synonym or id)

All Namespaces
4,779 of 207,564 terms · 96863 activities

FoodEx2 Classification
612 of 31,606 terms · 86,915 activities

AGROVOC Thesaurus
612 of 41,455 terms · 38,284 activities

HESTIA Glossary
374 of 36,050 terms · 81,221 activities

Geography
442 of 442 terms · 96,254 activities

ecoinvent Activities
1,745 of 33,786 terms · 1,338 activities

FoodOn Ontology
263 of 28,372 terms · 537 activities

Open Food Facts Categories
159 of 8,877 terms · 220 activities

UN Central Product Classification
134 of 4,583 terms · 37,196 activities

LanguaL Thesaurus
99 of 12,560 terms · 216 activities

GS1 Packaging Codes
22 of 168 terms · 73 activities

ecoinvent Activities

- Intermediate Exchanges
 - apple 1c819f15-1202-4fa9-8ef... in this project
 - pineapple 596ca18d-28a7-46db-bf1... in this project

eaternity-crop

- apple apple Fruit not in glossary in this project
- pineapple pineapple Fruit not in glossary in this project

FoodOn Ontology

- food material FOODON_00002403
 - food product FOODON_00001002
 - plant food product FOODON_00001015
 - vegetable food product FOODON_00001261
 - plant fruit food product FOODON_00001057
 - pomaceous fruit food product FOODON_00001158
 - apple food product FOODON_00001611
 - apple (whole or pieces) FOODON_03310788 14 narrower
 - apple FOODON_00002473 16 narrower in this project
 - apple (raw) FOODON_03301710 7 narrower
 - apple must FOODON_00002391
 - apple peel FOODON_03311637 1 narrower
 - apple sauce FOODON_03301126 2 narrower

See it. Compute it. Change it.

Graph-based view

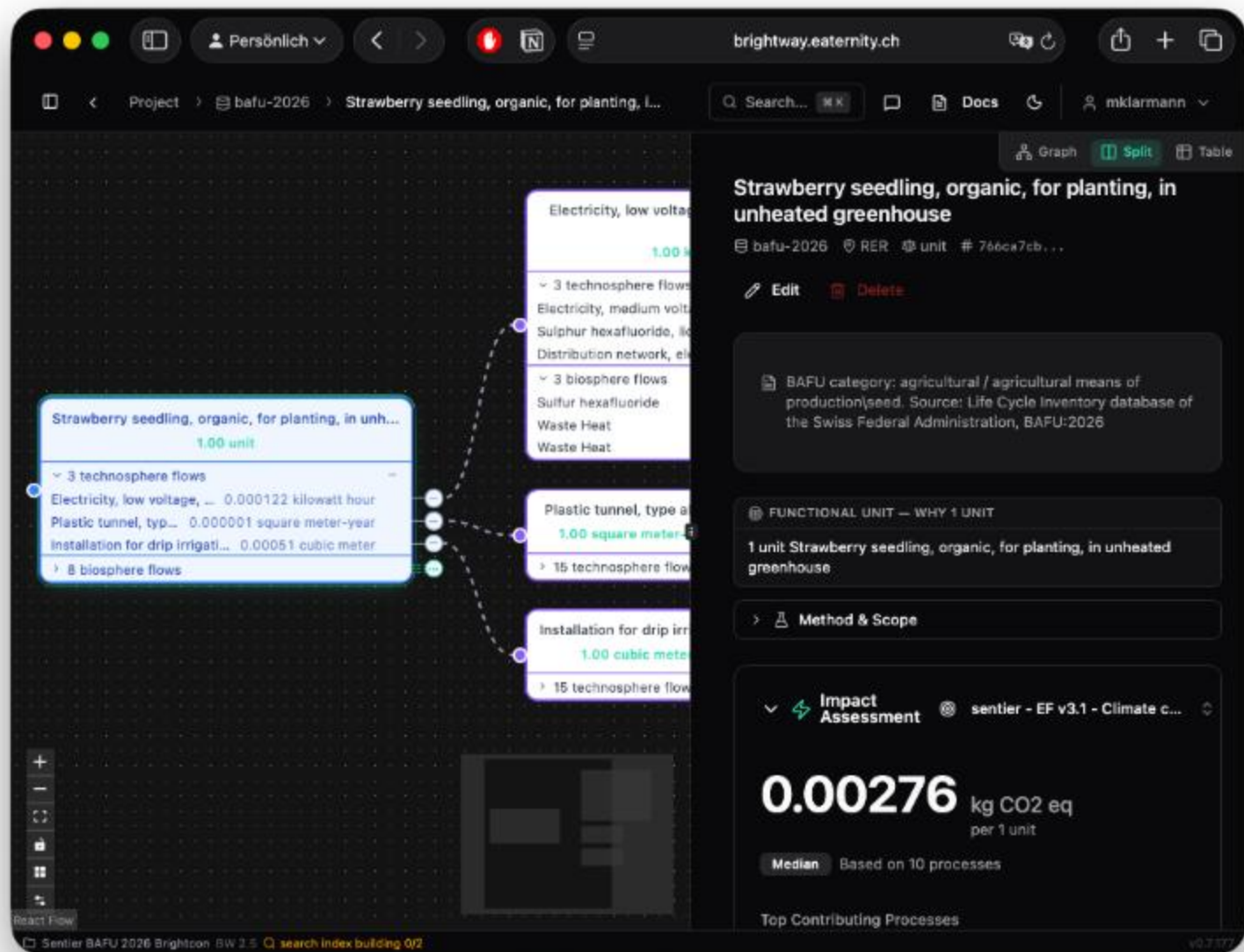
the supply chain drawn as it is; expand any node

Instant impact calculation

the score updates with the activity you open

Full information — and editable

sources, functional unit, method, every exchange

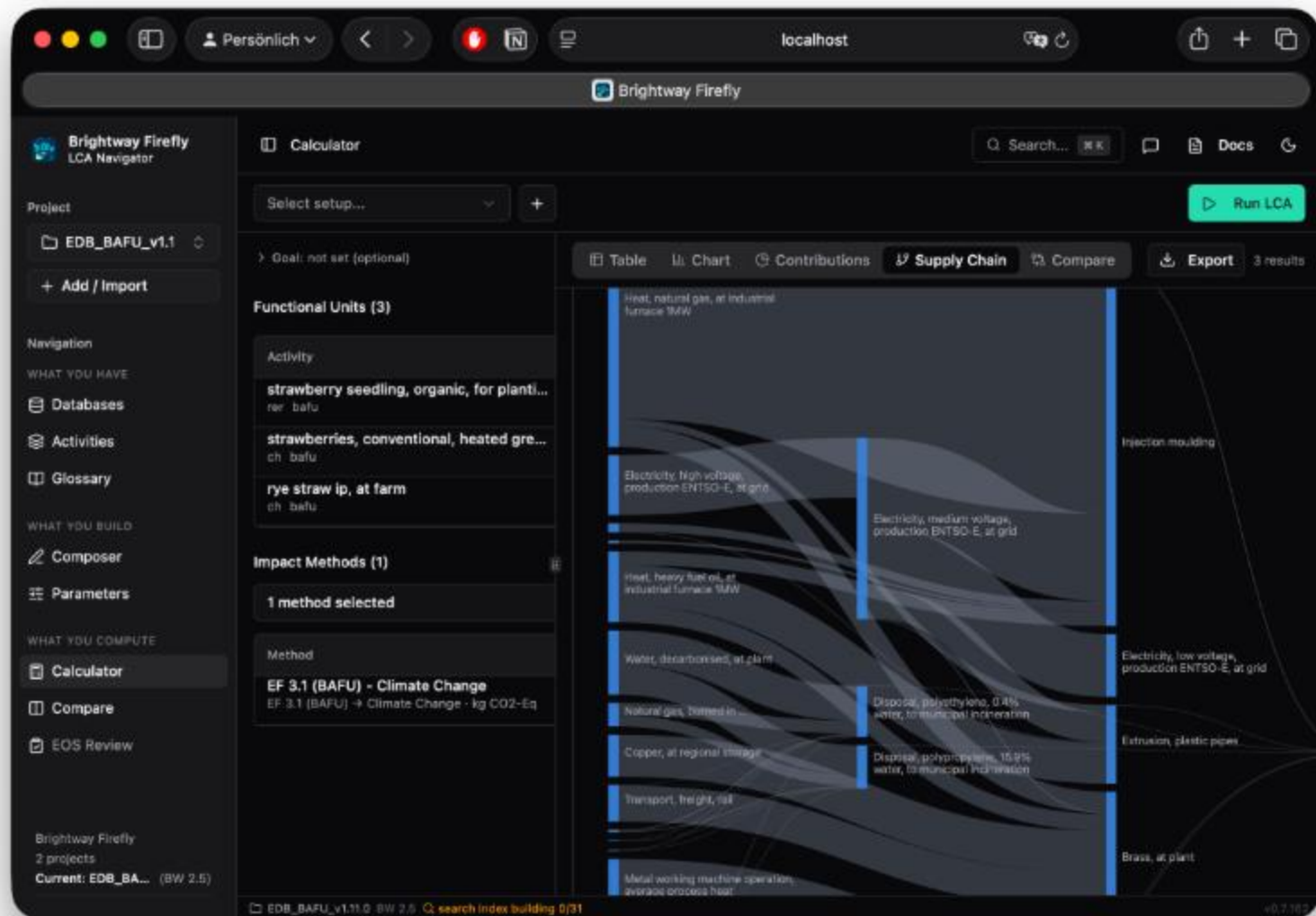


Run a calculation

Setup → functional units → methods → Run LCA

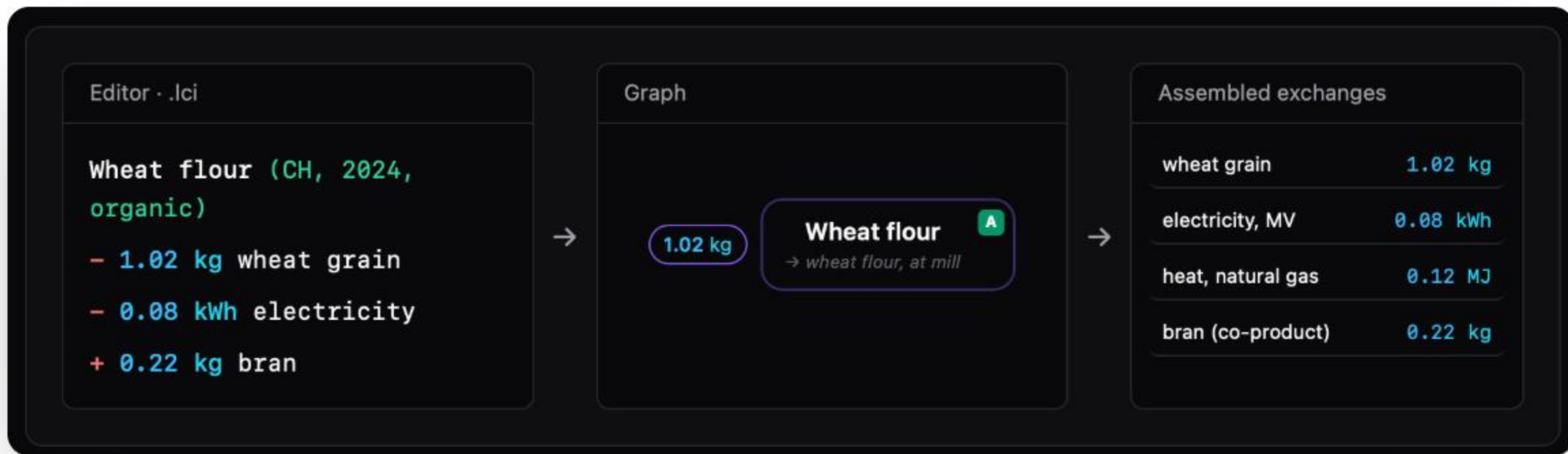
Results, charts, contribution, Sankey

Compare and Monte Carlo



The new Composer — a glossary-first editor

Write what you mean — the inventory assembles itself.



Text, not forms

one line per process and exchange — the graph and the exchanges are derived from it

Meaning, not keys

every name resolves through the glossary to the best inventory, with place, year and attributes in the line

Reproducible by design

the .lci file is the model: diff it, version it, share it — a better inventory re-scores the match

PART 2 · THE DATABASE

EDB — the Eaternity Database

A complete, fully transparent food inventory database. Every number can be opened, every model re-run.

19,289

inventories, transparent down to every exchange, source and model — nothing aggregated away

p5 · p95

an uncertainty band on every exchange; Monte Carlo where a model ran

country x year

most inventories are computed by models that take the country and the year as parameters

1,000

restaurants run on it today — in production at Eaternity

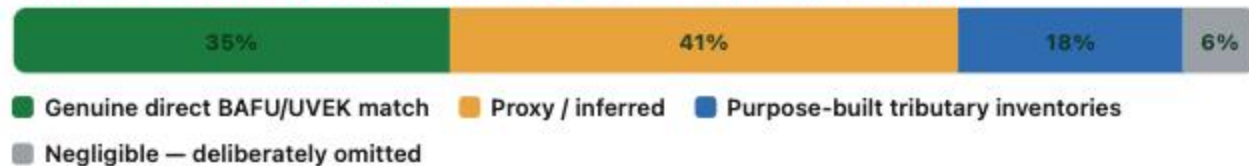
Mapping our database to BAFU/UEK: a third was missing

The Swiss BAFU/UEK database directly covers only about a third of the food background processes. The rest we built — hundreds of tributary inventories. Not a swap: an integration infrastructure.

DATABASE SIZE (ACTIVITIES) — REFERENCE LICENSED, BAFU/UEK PUBLIC



HOW THE BAFU VALUE OF THE 2,145 EDB BACKGROUND PROCESSES WAS DERIVED

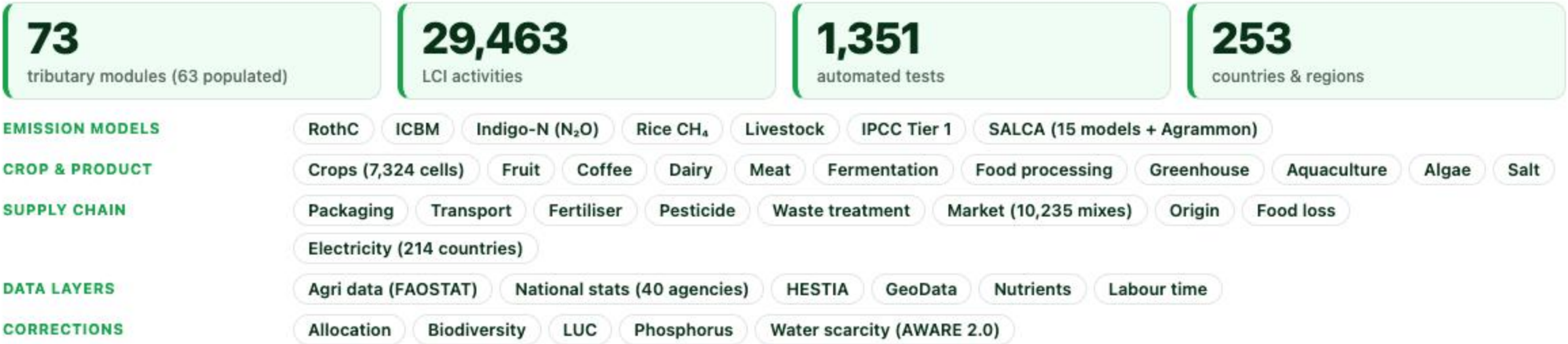


«Negligible» = deliberately omitted background contributions with no meaningful climate effect (e.g. machinery manufacture) — not missing inventories. Source: EDB base mapping, 2,145 background processes, 2026-06-17.

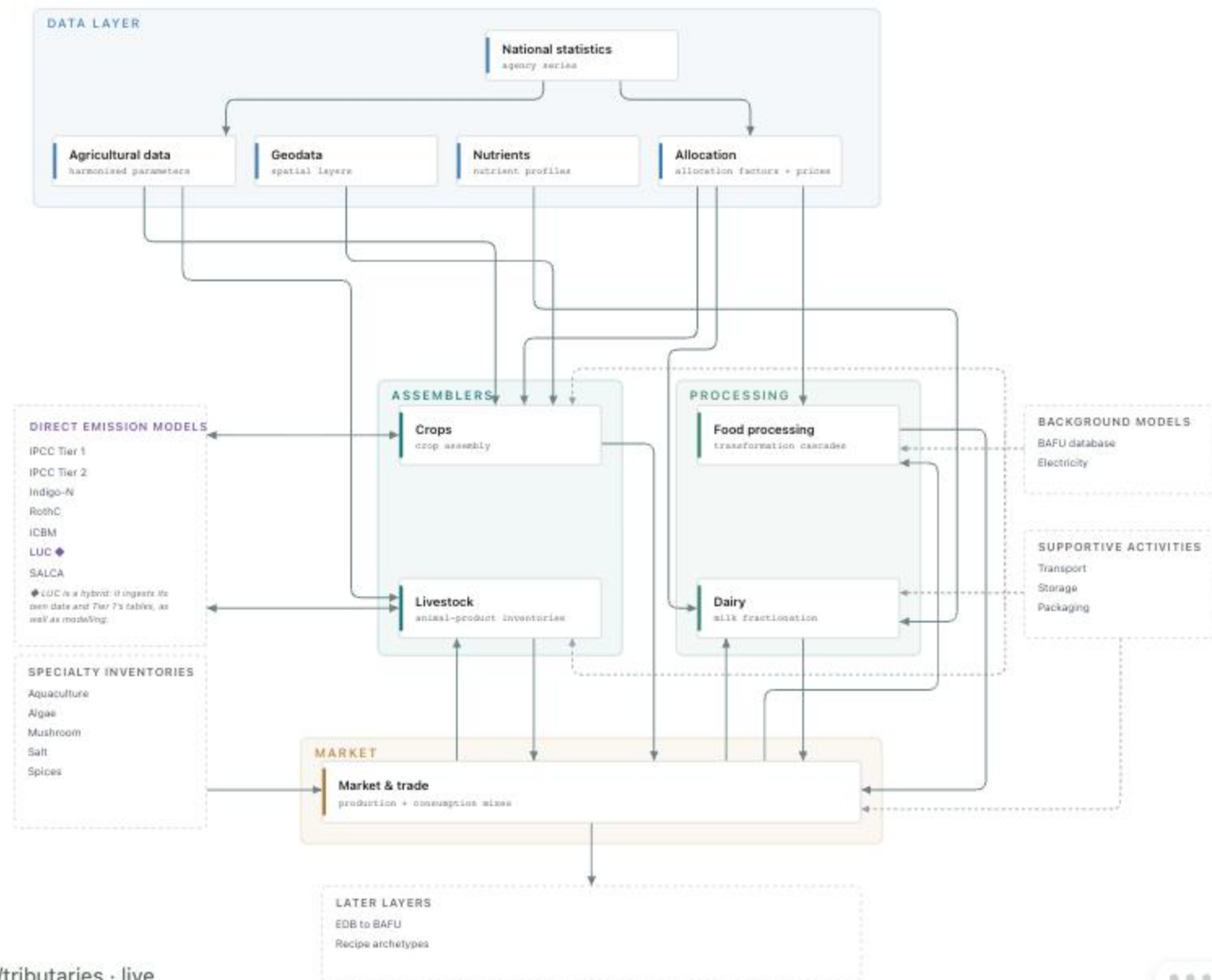


Models, not lookups — per country, per year, test-driven

73 tributary modules from national statistics and 3,700+ publications — per country, per year, test-driven.



How the database is composed



PART 2 · QUALITY THAT PREVENTS REGRESSIONS

One quality system, three questions

Evidence, scored

35 factors on every record

Regressions, caught

a ledger per artifact, checked daily

Hand-offs, guarded

~140 links between projects, drift-alarmed

lci-workbench-0d71e7.gitlab.io/#/quality

PART I

How good is the evidence in each record?

Every published record — inventory, mapping entry, contribution value — is graded by 35 factors on the evidence it actually carries.

PART II

Did anything regress over time?

Every scored artifact gets a committed ledger; its git history is the time series. A daily loop turns regressions into fixes or filed issues.

PART III

Do the parts still agree with each other?

Every hand-off between projects — factor tables, model inputs, query services — is declared, content-pinned, and drift-alarmed.

These are not three systems. One discovery pass finds every artifact, one scorer grades every record, one ledger format tracks every history, one watchdog guards all of it — the three questions are three surfaces of the same machinery.

35	4	50%	~140
registered factors (25 guards, 9 scores, 1 pedigree)	record kinds, each judged only on evidence it can carry	evidence floor — below it, no band is issued	cross-project hand-offs found by the architecture sweep

Re-derived from the data on every run — nothing is asserted by hand.



Project

EDB on BAFU...

Project: EDB_BAFU_ci

38 databases · 139.497 activities



Clone to edit

Search projects...

CONTEXT DATABASES

Sentier BAFU 2026

CI build

v0.1.0

a9dba75



Clone

EDB on BAFU/UEVK

published

v1.11.0

28eb771



Clone

✓ EDB on BAFU/UEVK

CI build

2026-09-24

28eb771



Clone

YOUR PROJECTS

Agribalyse test

20. Aug.

Agribalyse-v4.0-2026-AUG-13

20. Aug.

EDB-BAFU-AUGUST-11

20. Aug.

EDB_BAFU_published_mklarmann

today

Sentier BAFU 2026 Brightcon

today

EOS Review

Brightway Firefly

28 projects

Current: EDB_BAFU_ci

databases

— depends on — used by — mutual

_produc

capital_goods_

2

2 deps

construction_r

1

2 deps

Metals LCI

(Eaternity)

metals_lci

9

2 deps

storage_lci

3

2 deps

Livestock LCI

(Eaternity)

livestock_lci

73

7 deps

score-cache

28

2 deps

food_processi

217

16 deps

food_processi

300

13 deps

fuel_combusti

1

1 dep

EDB Remapped...

EDB_remapped_fl...

2.367

27

Open

Spices & Additives LC...

spices_additives_lci

24

5 deps

Aquaculture LCI (Eaternity)

aquaculture_lci

41

2 deps

Transport LCI (Eaternity)

transport_lci

21

1 dep

Algae LCI (Eaternity)

algae_lci

22

1 dep

cashew_lci

2

2 deps

Chemicals LCI (Eaternity)

chemicals_lci

9

2 deps

Coffee Green Bean...

coffee_green_bean

18

2 deps

Fermentation LCI (Eaternity)

fermentation_lci

60

2 deps

Irrigation LCI (Eaternity)

irrigation_lci

48

2 deps

mushroom_lci

3

2 deps

reefer_lci

ZHAW Proxy

Dairy LCI

Validation

Seed LCI

Two examples from the catalogue



How a tributary fills a concrete gap — using **Electricity** and **SALCA** as examples.

Electricity

grid mix by country & season

✓ electricity used in production

✓ fills an energy gap

SALCA

on-farm emissions

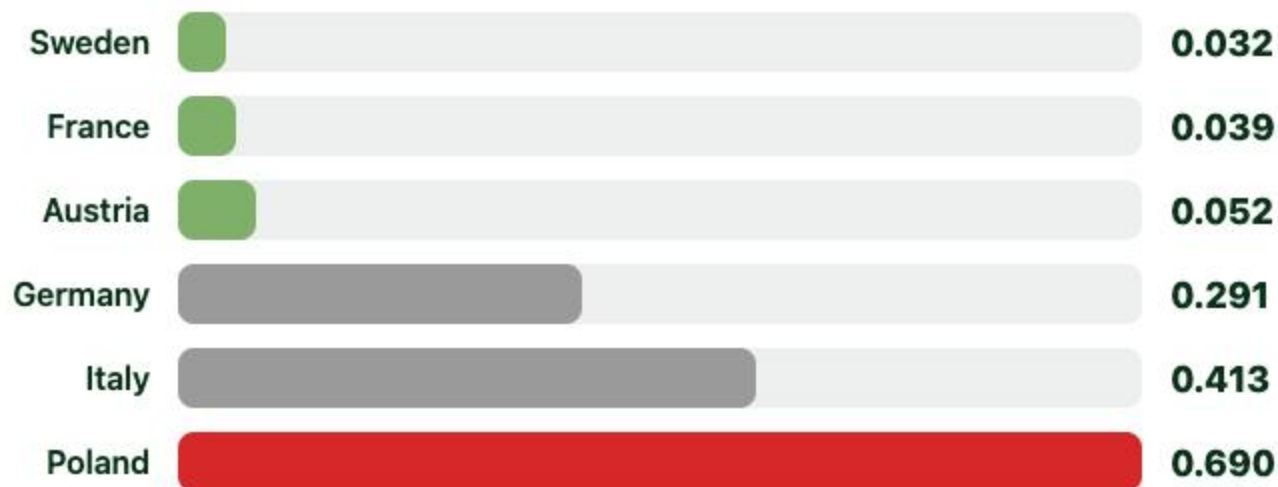
✓ N₂O, CH₄, soil carbon

✓ Agroscope method

Every tile is a tributary — click any to swap the example.

There is no «European electricity»

Which grid a product runs on decides its electricity footprint — roughly an order of magnitude across Europe. This tributary replaces one flat average with each country's measured mix, not a single value.



15+3

SALCA models re-implemented

48

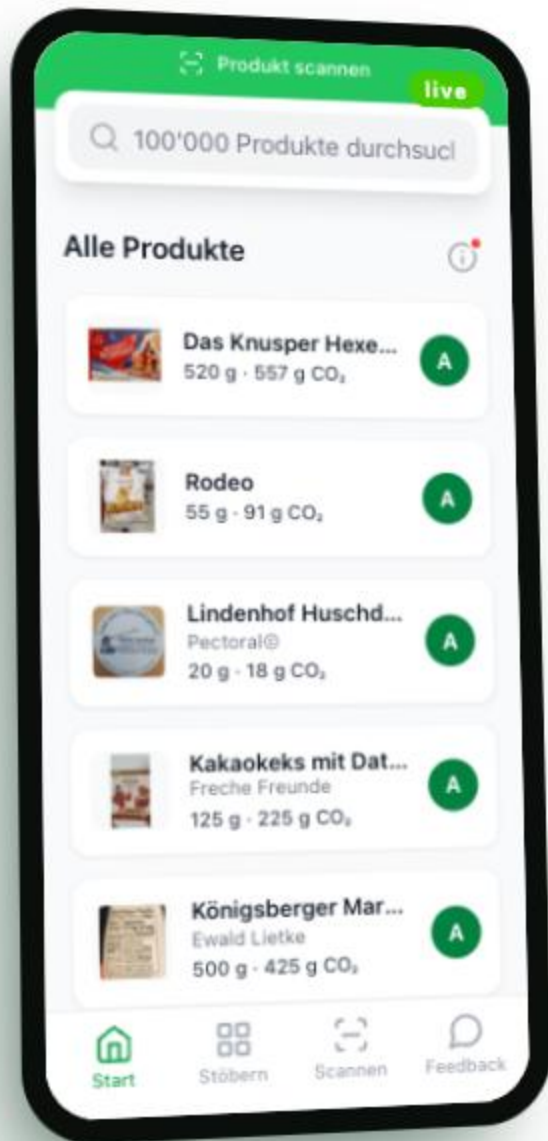
farm-gate activities, mechanism-resolved

SALCA MODEL

Built on Agroscope's SALCA method

SALCA is the established life-cycle assessment method of Agroscope's Life Cycle Assessment group — applied in research projects for years, and published on GitHub since June 2026. We re-implement the equation set in TypeScript, reconcile it against the official code, and wire it into our pipeline as a queryable, mechanism-resolved layer.

- ✓ Plus Agrammon NH_3 , MEXALCA, SALCA_{sustain}
- ✓ Field N_2O · NH_3 · NO_3 · field-P · CH_4 · soil-C — retrievable
- ✓ Reconciled against the official Agroscope code, anchored by tests



WHAT IT POWERS · FLIT

100,000 products, live

On this base we publish 100,000 European supermarket products — each with an A–E rating, a CO₂e value and a confidence score. Browse it here.

- 1 Search**
Any product by name or barcode
- 2 See the score**
A–E rating and CO₂e value
- 3 Review confidence**
Per-product transparency

flit.eaternity.org · live

ONE MORE THING

Free. Open. Yours.

All of it. Take it, build on it.

Apache-2.0

73 tributaries + the calculation engine

gitlab.com/eos-ici

CC-BY-4.0

19,289 inventories, the models, the results

no licence fee, no gatekeeper

AGPL-3.0

Firefly — the interface, the assistant, the MCP

gitlab.com/eaternity/firefly

Ask your project anything

 Claude + ANTHROPIC

Compare

Search...

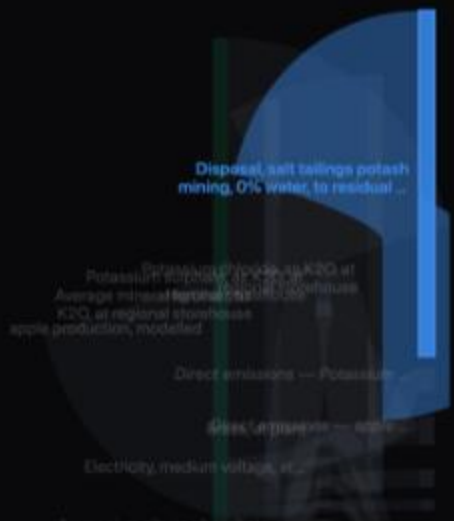
mklarmann

Logout

EDB-BAFU-AUGUST-11 · fruit_crops_lci

2.39 CTUe total — band width is the impact carried by each branch (dual-biosphere: base + BAUF methods summed)

28 nodes, 29 links (max depth: 3)



EDB-BAFU-AUGUST-11 · fruit_crops_lci

5.78 CTUe total — band width is the impact carried by each branch (dual-biosphere: base + BAUF methods summed)

33 nodes, 34 links (max depth: 3)



Assistant EDB-BAFU-AUGUST-11

80.394

tell me the contribution of pesticides to ecotoxicity to this apple

get_activity database=EDB code=3a9ef48dd91c09b1724917c416832e83_copy2

list_methods filter=ecotoxicity limit=40

calculate_lca database=EDB code=3a9ef48dd91c09b1724917c416832e83_copy2 method=["EF ...

contribution_analysis database=EDB code=3a9ef48dd91c09b1724917c416832e83_copy2 me...

contribution_analysis database=EDB code=3a9ef48dd91c09b1724917c416832e83_copy2 me...

I'll pull the activity and find the relevant ecotoxicity methods. Now the ecotoxicity calculation and a flow-level breakdown to isolate pesticides. ## Pesticides = 0.16 % of freshwater ecotoxicity

Total for 1 kg "market for apple" (EDB): 5.52 CTUe — method EF 3.1 (BAFU) — Ecotoxicity Freshwater, unit CTUe.

Pesticide active substances identified among the flow-level contributors:

Pesticide flow	Compartment	Inventory (kg/kg apple)	Impact (CTUe)	% of total
Carbendazim (fungicide)	soil	7.52e-9	0.00348	0.063 %
Carbofuran (insecticide)	soil	9.20e-9	0.00318	0.058 %
Monocrotophos (insecticide)	soil	5.62e-9	0.00149	0.027 %
Diffubenzuron (insecticide)	soil	4.60e-8	0.00092	0.017 %
Sum			≈ 0.0091	≈ 0.16 %

What actually drives ecotoxicity instead

Flow	Compartment	CTUe	%
Chloride	water	3.613	65.4 %

Claude Opus 5.5 — isolated, no internet

Brightway MCP · 22 tools

Delegates the busy work

Complex analysis on demand

22 / 23

N O W

Live demonstration

Everything you just saw is open: the interface, the database, the models, the assistant. Read it, audit it, reuse it.



brightway.eaternity.ch

- ✓ Firefly: brightway.eaternity.ch · gitlab.com/eaternity/firefly (AGPL-3.0)
- ✓ Database + 73 tributaries: gitlab.com/eos-lci (Apache-2.0 / CC-BY-4.0)
- ✓ Platform: lci-workbench-0d71e7.gitlab.io
- ✓ Flit: flit.eaternity.org